

Q1 - 24 January - Shift 1

For three positive integers p, q, r , $x^{pq^r} = y^{qr} = z^{p^r}$
 and $r = pq + 1$ such that $3, 3 \log_y x, 3 \log_z y, 7 \log_x z$
 are in A.P. with common difference $\frac{1}{2}$. Then
 $r - p - q$ is equal to

- (1) 2 (2) 6
 (3) 12 (4) -6

Space for your notes:

Q2 - 24 January - Shift 1

The 4th term of GP is 500 and its common ratio is
 $\frac{1}{m}$, $m \in \mathbb{N}$. Let S_n denote the sum of the first n
 terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$,
 then the number of possible values of m is

Space for your notes:

Q3 - 24 January - Shift 2

If $\frac{1^3 + 2^3 + 3^3 + \dots \text{upto } n \text{ terms}}{1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \dots \text{upto } n \text{ terms}} = \frac{9}{5}$, then
 the value of n is

Space for your notes:

Q4 - 25 January - Shift 1

Let A_1, A_2, A_3 be the three A.P. with the same common difference d and having their first terms as $A, A + 1, A + 2$, respectively. Let a, b, c be the $7^{\text{th}}, 9^{\text{th}}, 17^{\text{th}}$ terms of A_1, A_2, A_3 , respectively such

$$\text{that } \begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$$

If $a = 29$, then the sum of first 20 terms of an AP whose first term is $c - a - b$ and common difference is $\frac{d}{12}$, is equal to _____.

Space for your notes:

Q5 - 25 January - Shift 2

For the two positive numbers a, b , if a, b and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}, 10$ and $\frac{1}{b}$ are in an arithmetic progression, then, $16a + 12b$ is equal to _____.

Space for your notes:

Q6 - 29 January - Shift 1

Let $a_1, a_2, a_3, \dots$ be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1 a_9 + a_2 a_4 a_9 + a_5 + a_7$ is equal to _____.

Space for your notes:

Q7 - 29 January - Shift 2

Let $\{a_k\}$ and $\{b_k\}$, $k \in \mathbb{N}$, be two G.P.s with common ratio r_1 and r_2 respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k$, $k \in \mathbb{N}$.

If $c_2 = 5$ and $c_3 = \frac{13}{4}$ then $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$ is equal to _____.

Space for your notes:

Q8 - 29 January - Shift 2

Let $a_1 = b_1 = 1$ and $a_n = a_{n-1} + (n-1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^8 \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to _____.

Space for your notes:

Q9 - 30 January - Shift 1

If $a_n = \frac{-2}{4n^2 - 16n + 15}$, then $a_1 + a_2 + \dots + a_{25}$ is equal to :

Space for your notes:

- (1) $\frac{51}{144}$ (2) $\frac{49}{138}$
 (3) $\frac{50}{141}$ (4) $\frac{52}{147}$

Q10 - 30 January - Shift 1

Let
$$\sum_{n=0}^{\infty} \frac{n^3 ((2n)!) + (2n-1)(n!)}{(n!)((2n)!)} = ae + \frac{b}{e} + c,$$

where $a, b, c \in \mathbb{Z}$ and $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. Then $a^2 - b + c$ is equal to _____.

Space for your notes:

Q11 - 30 January - Shift 2

Let $a, b, c > 1$, $a^{\frac{1}{3}}$, $b^{\frac{1}{3}}$ and $c^{\frac{1}{3}}$ be in A.P., and $\log_a b$, $\log_b c$ and $\log_c a$ be in G.P. If the sum of first 20 terms of an A.P., whose first term is $\frac{a+4b+c}{3}$

and the common difference is $\frac{a-8b+c}{10}$ is -444 ,

then abc is equal to

- (1) 343 (2) 216
(3) $\frac{343}{8}$ (4) $\frac{125}{8}$

Space for your notes:

Q12 - 30 January - Shift 2

The parabolas : $ax^2 + 2bx + cy = 0$ and $dx^2 + 2ex + fy = 0$ intersect on the line $y = 1$. If a, b, c, d, e, f are positive real numbers and a, b, c are in G.P., then

- (1) d, e, f are in A.P. (2) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

- (3) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (4) d, e, f are in G.P.

Space for your notes:

Q13 - 30 January - Shift 2

The 8th common term of the series

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$$S_1 = 3 + 7 + 11 + 15 + 19 + \dots,$$

$$S_2 = 1 + 6 + 11 + 16 + 21 + \dots$$

is _____.

Q14 - 31 January - Shift 1

Let $y = f(x)$ represent a parabola with focus

$$\left(-\frac{1}{2}, 0\right) \text{ and directrix } y = -\frac{1}{2}.$$

Space for your notes:

Then

$$S = \left\{ x \in \mathbb{R} : \tan^{-1} \left(\sqrt{f(x)} + \sin^{-1} \left(\sqrt{f(x)+1} \right) \right) = \frac{\pi}{2} \right\} :$$

(1) contains exactly two elements

(2) contains exactly one element

(3) is an infinite set

(4) is an empty set

Q15 - 31 January - Shift 1

Let $a_1, a_2, \dots, a_n$ be in A.P. If $a_5 = 2a_7$ and

Space for your notes:

$a_{11} = 18$, then

$$12 \left(\frac{1}{\sqrt{a_{10}} + \sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}} + \sqrt{a_{12}}} + \dots + \frac{1}{\sqrt{a_{17}} + \sqrt{a_{18}}} \right)$$

is equal to _____.

Q16 - 31 January - Shift 2

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Let $a_1, a_2, a_3, \dots$ be an A.P. If $a_7 = 3$, the product $a_1 a_4$ is minimum and the sum of its first n terms is zero, then $n! - 4a_{n(n+2)}$ is equal to :

Space for your notes:

(1) 24

(2) $\frac{33}{4}$

(3) $\frac{381}{4}$

(4) 9

Q17 - 31 January - Shift 2

The sum $1^2 - 2 \cdot 3^2 + 3 \cdot 5^2 - 4 \cdot 7^2 + 5 \cdot 9^2 - \dots + 15 \cdot 29^2$ is _____.

Space for your notes:

Q18 - 01 February - Shift 1

The sum to 10 terms of the series

Space for your notes:

$$\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots \text{ is:-}$$

(1) $\frac{59}{111}$ (2) $\frac{55}{111}$

(3) $\frac{56}{111}$ (4) $\frac{58}{111}$

Q19 - 01 February - Shift 1

Let $a_1 = 8, a_2, a_3, \dots, a_n$ be an A.P. If the sum of its first four terms is 50 and the sum of its last four terms is 170, then the product of its middle two terms is _____.

Space for your notes:

Q20 - 01 February - Shift 1

The number of 3-digit numbers, that are divisible by either 2 or 3 but not divisible by 7 is _____.

Space for your notes:

Q21 - 01 February - Shift 2

Which of the following statements is a tautology ?

Space for your notes:

- (1) $p \rightarrow (p \wedge (p \rightarrow q))$
- (2) $(p \wedge q) \rightarrow (\sim(p) \rightarrow q)$
- (3) $(p \wedge (p \rightarrow q)) \rightarrow \sim q$
- (4) $p \vee (p \wedge q)$

Q22 - 01 February - Shift 2

The sum of the common terms of the following three arithmetic progressions.

Space for your notes:

3, 7, 11, 15,, 399,

2, 5, 8, 11,, 359 and

2, 7, 12, 17,, 197, is equal to _____.

Answer Key

Answer Key      

(As per Official NTA Key released on 2 Feb)

Q1 (1)  **Q2 (12)**  **Q3 (5)**  **Q4 (495)**  

Q5 (3)  **Q6 (60)**  **Q7 (9)**  **Q8 (461)**  

Q9 (3)  **Q10 (26)**  **Q11 (2)**  **Q12 (3)**  

Q13 (151)  **Q14 (1)**  **Q15 (8)**  **Q16 (1)**  

Q17 (6952) **Q18 (2)** **Q19 (754)** **Q20 (514)**

Q21 (2) **Q22 (321)**

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 Q1 (1)					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

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$$pq^2 = \log_x \lambda$$

$$qr = \log_y \lambda$$

$$p^2 r = \log_z \lambda$$

$$\log_y x = \frac{qr}{pq^2} = \frac{r}{pq} \dots\dots\dots(1)$$

$$\log_x z = \frac{pq^2}{p^2 r} = \frac{q^2}{pr} \dots\dots\dots(2)$$

$$\log_z y = \frac{p^2 r}{qr} = \frac{p^2}{q} \dots\dots\dots(3)$$

$$3, \frac{3r}{pq}, \frac{3p^2}{q}, \frac{7q^2}{pr} \text{ in A.P}$$

$$\frac{3r}{pq} - 3 = \frac{1}{2}$$

$$r = \frac{7}{6} pq \dots\dots\dots(4)$$

$$r = pq + 1$$

$$pq = 6 \dots\dots\dots(5)$$

$$r = 7 \dots\dots\dots(6)$$

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Q2 (12)

 $T_4 = 500$ where a = first term,

 r = common ratio = $\frac{1}{m}$, $m \in \mathbb{N}$

$$ar^3 = 500$$

$$\frac{a}{m^3} = 500$$

$$S_n - S_{n-1} = ar^{n-1}$$

$$S_6 > S_5 + 1 \quad \text{and} \quad S_7 - S_6 < \frac{1}{2}$$

$$S_6 - S_5 > 1 \quad \frac{a}{m^6} < \frac{1}{2}$$

$$ar^5 > 1 \quad m^3 > 10^3$$

$$\frac{500}{m^2} > 1 \quad m > 10 \dots\dots(2)$$

$$m^2 < 500 \dots\dots(1)$$

From (1) and (2)

$$m = 11, 12, 13, \dots\dots\dots, 22$$

So number of possible values of m is 12

Q3 (5)

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \dots + n \text{ terms} =$$

$$\sum_{r=1}^n r(2r+1) = \sum_{r=1}^n (2r^2 + r)$$

$$= \frac{2 \cdot n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{6} (2(2n+1) + 3)$$

$$= \frac{n(n+1)}{2} \times \frac{(4n+5)}{3}$$

$$= \frac{\frac{n^2(n+1)^2}{4}}{\frac{n(n+1)}{2} \times \frac{(4n+5)}{3}} = \frac{9}{5}$$

$$\Rightarrow \frac{5n(n+1)}{2} = \frac{9(4n+5)}{3}$$

$$\Rightarrow 15n(n+1) = 18(4n+5)$$

$$\Rightarrow 15n^2 + 15n = 72n + 90$$

$$\Rightarrow 15n^2 - 57n - 90 = 0 \Rightarrow 5n^2 - 19n - 30 = 0$$

$$\Rightarrow (n-5)(5n+6) = 0$$

$$\Rightarrow n = \frac{-6}{5} \text{ or } 5$$

$$\Rightarrow n = 5.$$

Q4 (495)

$$\begin{vmatrix} A + 6d & 7 & 1 \\ 2(A + 1 + 8d) & 17 & 1 \\ A + 2 + 16d & 17 & 1 \end{vmatrix} + 70 = 0$$

$$\Rightarrow A = -7 \text{ and } d = 6$$

$$\therefore c - a - b = 20$$

$$S_{20} = 495$$

Q5 (3)

$$a, b, \frac{1}{18} \rightarrow \text{GP}$$

$$\frac{a}{18} = b^2 \dots\dots (i)$$

$$\frac{1}{a}, 10, \frac{1}{b} \rightarrow \text{AP}$$

$$\frac{1}{a} + \frac{1}{b} = 20$$

$$\Rightarrow a + b = 20ab, \text{ from eq. (i) ; we get}$$

$$\Rightarrow 18b^2 + b = 360b^3$$

$$\Rightarrow 360b^2 - 18b - 1 = 0 \quad \{\because b \neq 0\}$$

$$\Rightarrow b = \frac{18 \pm \sqrt{324 + 1440}}{720}$$

$$\Rightarrow b = \frac{18 + \sqrt{1764}}{720} \quad \{\because b > 0\}$$

$$\Rightarrow b = \frac{1}{12}$$

$$\Rightarrow a = 18 \times \frac{1}{144} = \frac{1}{8}$$

$$\text{Now, } 16a + 12b = 16 \times \frac{1}{8} + 12 \times \frac{1}{12} = 3$$

Q6 (60)

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$$a_4 \cdot a_6 = 9 \Rightarrow (a_5)^2 = 9 \Rightarrow a_5 = 3$$

$$\& a_5 + a_7 = 24 \Rightarrow a_5 + a_5 r^2 = 24 \Rightarrow (1 + r^2) = 8 \Rightarrow r = \sqrt{7}$$

$$\Rightarrow a = \frac{3}{49}$$

$$\Rightarrow a_1 a_9 + a_2 a_4 a_9 + a_5 + a_7 = 9 + 27 + 3 + 21 = 60$$

Q7 (9)

Given that

$$c_k = a_k + b_k \text{ and } a_1 = b_1 = 4$$

$$\text{also } a_2 = 4r_1 \quad a_3 = 4r_1^2$$

$$b_2 = 4r_2 \quad b_3 = 4r_2^2$$

$$\text{Now } c_2 = a_2 + b_2 = 5 \text{ and } c_3 = a_3 + b_3 = \frac{13}{4}$$

$$\Rightarrow r_1 + r_2 = \frac{5}{4} \text{ and } r_1^2 + r_2^2 = \frac{13}{16}$$

$$\text{Hence } r_1 r_2 = \frac{3}{8} \text{ which gives } r_1 = \frac{1}{2} \text{ \& } r_2 = \frac{3}{4}$$

$$\sum_{k=1}^{\infty} c_k = (12a_6 + 8b_4)$$

$$= \frac{4}{1-r_1} + \frac{4}{1-r_2} = \left(\frac{48}{32} + \frac{27}{2} \right)$$

$$= 24 - 15 = 9$$

Q8 (461)

Questions with Solutions

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$$\text{As, } S = \frac{a_1}{2} + \frac{a_2}{2^2} + \dots + \frac{a_9}{2^9} + \frac{a_{10}}{2^{10}}$$

$$\Rightarrow \frac{S}{2} = \frac{a_1}{2^2} + \frac{a_2}{2^3} + \dots + \frac{a_9}{2^{10}} + \frac{a_{10}}{2^{11}}$$

subtracting

$$\Rightarrow \frac{S}{2} = \frac{a_1}{2} + \left(\frac{a_1}{2^2} + \frac{a_2}{2^3} + \dots + \frac{a_9}{2^{10}} \right) - \frac{a_{10}}{2^{11}}$$

$$\Rightarrow S = a_1 - \frac{a_{10}}{2^{10}} + \left(\frac{a_1}{2} + \frac{a_2}{2^2} + \dots + \frac{a_9}{2^9} \right)$$

$$\Rightarrow \frac{S}{2} = \frac{a_1}{2} - \frac{a_{10}}{2^{11}} + \left(\frac{a_1}{2^2} + \frac{a_2}{2^3} + \dots + \frac{a_9}{2^{10}} \right)$$

subtracting

$$\Rightarrow \frac{S}{2} = \frac{a_1}{2} - \frac{a_{10}}{2^{11}} + \left(\frac{a_1}{2} - \frac{a_9}{2^{10}} \right) + \left(\frac{1}{2^2} + \frac{2}{2^3} + \dots + \frac{8}{2^9} \right)$$

$$\Rightarrow \frac{S}{2} = \frac{a_1 + b_1}{2} - \frac{(b_{10} + 2a_9)}{2^{11}} + \frac{T}{4}$$

$$\Rightarrow 2S = 2(a_1 + b_1) - \frac{(b_{10} + 2a_9)}{2^9} + T$$

$$\Rightarrow 2^7 (2S - T) = 2^8 (a_1 + b_1) - \frac{(b_{10} + 2a_9)}{4}$$

$$\text{Given } a_n - a_{n-1} = n - 1,$$

$$\therefore a_2 - a_1 = 1$$

$$a_3 - a_2 = 2$$

⋮

$$a_9 - a_8 = 8$$

$$a_9 - a_1 = 1 + 2 + \dots + 8 = 36$$

$$\Rightarrow a_9 = 37 \quad (a_1 = 1)$$

$$\text{Also, } b_n - b_{n-1} = a_{n-1}$$

$$\therefore b_{10} - b_1 = a_1 + a_2 + \dots + a_9$$

$$= 1 + 2 + 4 + 7 + 11 + 16 + 22 + 29 + 37$$

$$\Rightarrow b_{10} = 130 \quad (\text{As } b_1 = 1)$$

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Q9 (3)

Option (3)

If $a_n = \frac{-2}{4n^2 - 16n + 15}$ then $a_1 + a_2 + \dots + a_{25}$

$$\Rightarrow \sum_{n=1}^{25} a_n = \sum \frac{-2}{4n^2 - 16n + 15}$$

$$= \sum \frac{-2}{4n^2 - 6n - 10n + 15}$$

$$= \sum \frac{-2}{2n(2n-3) - 5(2n-3)}$$

$$= \sum \frac{-2}{(2n-3)(2n-5)}$$

$$= \sum \frac{1}{2n-3} - \frac{1}{2n-5}$$

$$= \frac{1}{47} - \frac{1}{(-3)}$$

$$= \frac{50}{141}$$

Q10 (26)

$$\begin{aligned}
 & \sum_{n=0}^{\infty} \frac{n^3 ((2n)!) + (2n-1)(n!)}{(n!)((2n)!)} \\
 &= \sum_{n=0}^{\infty} \frac{1}{(n-3)!} + \sum_{n=0}^{\infty} \frac{3}{(n-2)!} \\
 &+ \sum_{n=0}^{\infty} \frac{1}{(n-1)!} + \sum_{n=0}^{\infty} \frac{1}{(2n-1)!} - \sum_{n=0}^{\infty} \frac{1}{(2n)!} \\
 &= e + 3e + e + \frac{1}{2} \left(e - \frac{1}{e} \right) - \frac{1}{2} \left(e + \frac{1}{e} \right) \\
 &= 5e - \frac{1}{e}
 \end{aligned}$$

$$a^2 - b + c = 26$$

Q11 (2)

As a^3, b^3, c^3 be in A.P. $\rightarrow \boxed{a^3 + c^3 = 2b^3}$ (1)

$\log_a^b, \log_c^a, \log_b^c$ are in G.P.

$$\therefore \frac{\log b}{\log a} \cdot \frac{\log c}{\log b} = \left(\frac{\log a}{\log c} \right)^2$$

$$\therefore (\log a)^3 = (\log c)^3 \Rightarrow \boxed{a = c}$$
 (2)

From (1) and (2)

$$\boxed{a = b = c}$$

$$T_1 = \frac{a + 4b + c}{3} = 2a; d = \frac{a - 8b + c}{10} = \frac{-6a}{10} = -\frac{3}{5}a$$

$$\therefore S_{20} = \frac{20}{2} \left[4a + 19 \left(-\frac{3}{5}a \right) \right]$$

$$= 10 \left[\frac{20a - 57a}{5} \right]$$

$$= -74a$$

$$\therefore -74a = -444 \Rightarrow \boxed{a = 6}$$

$$\therefore abc = 6^3 = 216$$

Q12 (3)

$$ax^2 + 2bx + c = 0$$

$$\Rightarrow ax^2 + 2\sqrt{ac}x + c = 0 (\because b^2 = ac)$$

$$\Rightarrow (x\sqrt{a} + \sqrt{c})^2 = 0$$

$$x^2 + \frac{\sqrt{c}}{\sqrt{a}}x + \frac{\sqrt{c}}{\sqrt{a}} = 0 \dots (1)$$

$$\text{Now, } dx^2 + 2ex + f = 0$$

$$\Rightarrow d\left(\frac{c}{a}\right) + 2e\left[-\frac{\sqrt{c}}{\sqrt{a}}\right] + f = 0$$

$$\Rightarrow \frac{dc}{a} + f = 2e\sqrt{\frac{c}{a}}$$

$$\Rightarrow \frac{d}{a} + \frac{f}{c} = 2e\sqrt{\frac{1}{ac}}$$

$$\Rightarrow \frac{d}{a} + \frac{f}{c} = \frac{2e}{b} \left[\text{as } b = \sqrt{ac} \right]$$

$$\therefore \frac{d}{a}, \frac{e}{b}, \frac{f}{c} \text{ are in A.P.}$$

Q13 (151)

$$T_8 = 11 + (8 - 1) \times 20$$

$$= 11 + 140 = 151$$

Q14 (1)

$$\left(x + \frac{1}{2}\right)^2 = \left(y + \frac{1}{4}\right)$$

$$y = (x^2 + x)$$

$$\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$$

$$0 \leq x^2 + x + 1 \leq 1$$

$$x^2 + x \leq 0 \quad \dots(1)$$

$$\text{Also } x^2 + x \geq 0 \quad \dots(2)$$

$$\therefore x^2 + x = 0 \Rightarrow x = 0, -1$$

S contains 2 element.

Q15 (8)

$$2a_7 = a_5 \text{ (given)}$$

$$2(a_1 + 6d) = a_1 + 4d$$

$$a_1 + 8d = 0 \quad \dots\dots(1)$$

$$a_1 + 10d = 18 \quad \dots\dots(2)$$

By (1) and (2) we get $a_1 = -72$, $d = 9$

$$a_{18} = a_1 + 17d = -72 + 153 = 81$$

$$a_{10} = a_1 + 9d = 9$$

$$12 \left(\frac{\sqrt{a_{11}} - \sqrt{a_{10}}}{d} + \frac{\sqrt{a_{12}} - \sqrt{a_{11}}}{d} + \dots\dots \frac{\sqrt{a_{18}} - \sqrt{a_{17}}}{d} \right)$$

$$12 \left(\frac{\sqrt{a_{18}} - \sqrt{a_{10}}}{d} \right) = \frac{12(9-3)}{9} = \frac{12 \times 6}{6} = 8$$

Q16 (1)

$$a+6d=3, \dots\dots\dots(1)$$

$$Z = a(a+3d) \\ = (3-6d)(3-3d)$$

$$= 18d^2 - 27d + 9$$

Differentiating with respect to d

$$\Rightarrow 36d - 27 = 0$$

$$\Rightarrow d = \frac{3}{4}, \text{ from (1) } a = \frac{-3}{2}, (Z = \text{minimum})$$

$$\text{Now, } S_n = \frac{n}{2} \left(-3 + (n-1) \frac{3}{4} \right) = 0$$

$$\Rightarrow n = 5$$

Now,

$$n! - 4a_{n(n+2)} = 120 - 4(a_{35})$$

$$= 120 - 4 \left(a + (35-1)d \right)$$

$$= 120 - 4 \left(\frac{-3}{2} + 34 \cdot \left(\frac{3}{4} \right) \right)$$

$$= 120 - 4 \left(\frac{-6 + 102}{4} \right)$$

$$= 120 - 96 = 24$$

Q17 (6952)

Separating odd placed and even placed terms we get

$$S = (1.1^2 + 3.5^2 + \dots + 15.(29)^2) - (2.3^2 + 4.7^2 + \dots + 14.(27)^2)$$

$$S = \sum_{n=1}^8 (2n-1)(4n-3)^2 - \sum_{n=1}^7 (2n)(4n-1)^2$$

Applying summation formula we get

$$= 29856 - 22904 = 6952$$

Q18 (2)

$$T_r = \frac{(r^2 + r + 1) - (r^2 - r + 1)}{2(r^4 + r^2 + 1)}$$

$$\Rightarrow T_r = \frac{1}{2} \left[\frac{1}{r^2 - r + 1} - \frac{1}{r^2 + r + 1} \right]$$

$$T_1 = \frac{1}{2} \left[\frac{1}{1} - \frac{1}{3} \right]$$

$$T_2 = \frac{1}{2} \left[\frac{1}{3} - \frac{1}{7} \right]$$

$$T_3 = \frac{1}{2} \left[\frac{1}{7} - \frac{1}{13} \right]$$

$$\vdots$$

$$T_{10} = \frac{1}{2} \left[\frac{1}{91} - \frac{1}{111} \right]$$

$$\Rightarrow \sum_{r=1}^{10} T_r = \frac{1}{2} \left[1 - \frac{1}{111} \right] = \frac{55}{111}$$

Q19 (754)

$$a_1 + a_2 + a_3 + a_4 = 50$$

$$\Rightarrow 32 + 6d = 50$$

$$\Rightarrow d = 3$$

$$\text{and, } a_{n-3} + a_{n-2} + a_{n-1} + a_n = 170$$

$$\Rightarrow 32 + (4n - 10).3 = 170$$

$$\Rightarrow n = 14$$

$$a_7 = 26, a_8 = 29$$

$$\Rightarrow a_7 \cdot a_8 = 754$$

Q20 (514)

Divisible by 2 $\rightarrow$ 450

Divisible by 3 $\rightarrow$ 300

Divisible by 7 $\rightarrow$ 128

Divisible by 2 & 7 $\rightarrow$ 64

Divisible by 3 & 7 $\rightarrow$ 43

Divisible by 2 & 3 $\rightarrow$ 150

Divisible by 2, 3 & 7 $\rightarrow$ 21

$\therefore$ Total numbers = $450 + 300 - 150 - 64 - 43 +$

$21 = 514$

Q21 (2)

$$(i) p \rightarrow (p \wedge (p \rightarrow q))$$

$$(\sim p) \vee (p \wedge (\sim p \vee q))$$

$$(\sim p) \vee (p \wedge q)$$

$$\sim p \vee (p \wedge q) = (\sim p \vee p) \wedge (\sim p \vee q)$$

$$= \sim p \vee q$$

$$(ii) (p \wedge q) \rightarrow (\sim p \rightarrow q)$$

$$\sim(p \wedge q) \vee (p \vee q) = t$$

$$\{a, b, d\} \vee \{a, b, c\} = V$$

Tautology

$$(iii) (p \wedge (p \rightarrow q)) \rightarrow \sim q$$

$$\sim(p \wedge (\sim p \vee q)) \vee \sim q = \sim(p \wedge q) \vee \sim q = \sim p \vee \sim q$$

Not tautology

$$(iv) p \vee (p \wedge q) = p$$

Not tautology.

Q22 (321)

$$3, 7, 11, 15, \dots, 399 \quad d_1 = 4$$

$$2, 5, 8, 11, \dots, 359 \quad d_2 = 3$$

$$2, 7, 12, 17, \dots, 197 \quad d_3 = 5$$

$$\text{LCM}(d_1, d_2, d_3) = 60$$

Common terms are 47, 107, 167

$$\text{Sum} = 321$$